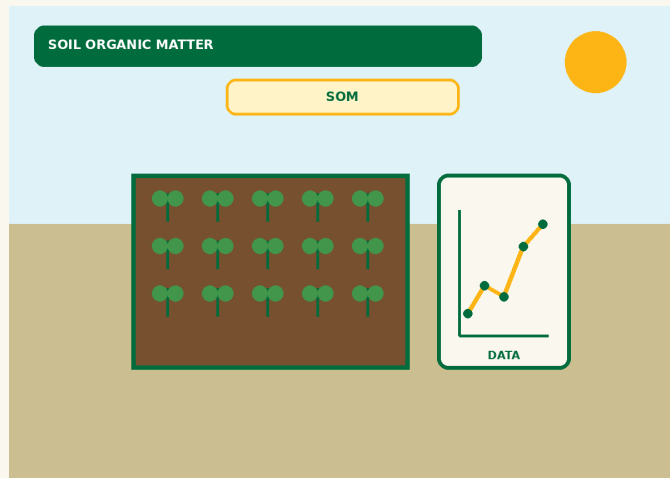




SOM

SOIL ORGANIC MATTER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 721 • SOIL ORGANIC MATTER

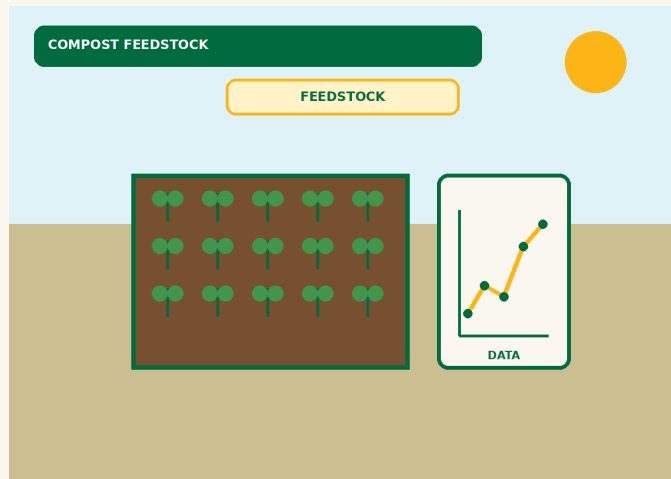
CONCEPT

Soil organic matter consists of carbon-containing materials at different stages of decomposition and influences many soil functions.

INVESTIGATION TASK

Identify management practices that can affect SOM.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FEEDSTOCK

COMPOST FEEDSTOCK

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 722 • COMPOST FEEDSTOCK

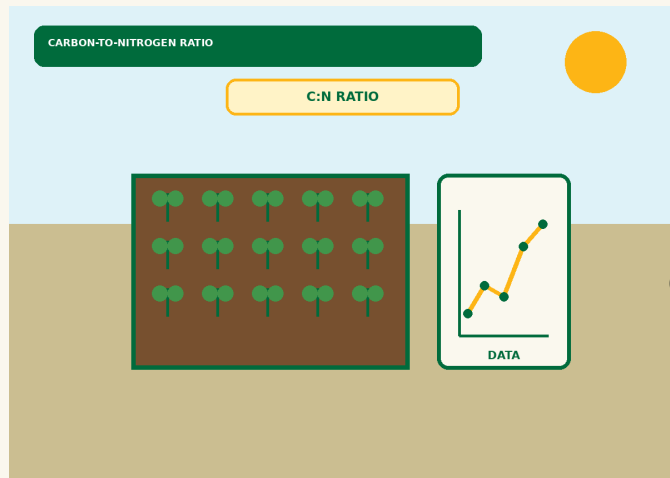
CONCEPT

Feedstock selection influences carbon, nitrogen, moisture, structure, contaminants, and decomposition rate.

INVESTIGATION TASK

Evaluate a proposed feedstock mixture.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 723 • CARBON-TO-NITROGEN RATIO

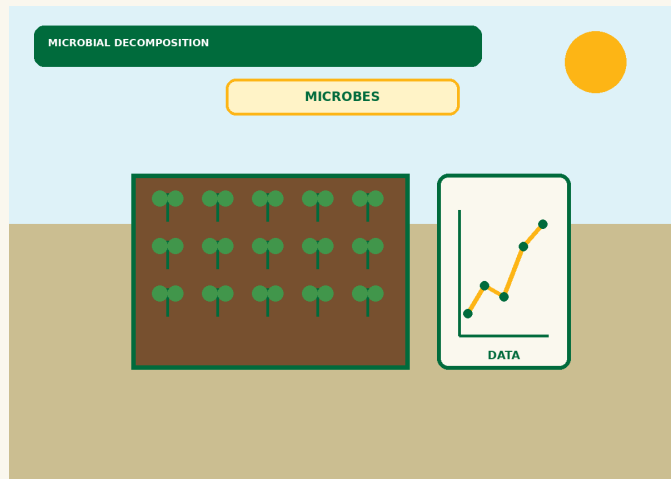
CONCEPT

The carbon-to-nitrogen ratio describes the relative amounts of carbon and nitrogen in composting materials and influences decomposition.

INVESTIGATION TASK

Interpret simplified C:N information.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MICROBES

MICROBIAL DECOMPOSITION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 724 • MICROBIAL DECOMPOSITION

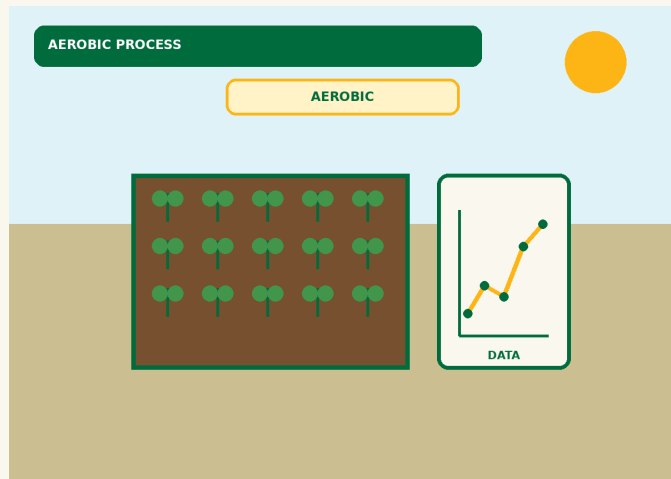
CONCEPT

Microorganisms transform organic materials while using carbon, nutrients, water, and oxygen.

INVESTIGATION TASK

Explain why microbial activity can raise pile temperature.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



AEROBIC

AEROBIC PROCESS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 725 • AEROBIC PROCESS

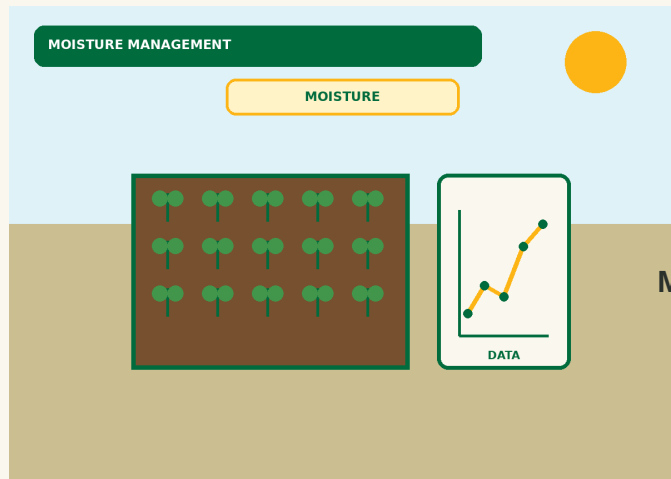
CONCEPT

Aerobic decomposition occurs in the presence of oxygen and is supported by adequate pore space and turning when needed.

INVESTIGATION TASK

Identify signs of poor aeration.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MOISTURE

MOISTURE MANAGEMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 726 • MOISTURE MANAGEMENT

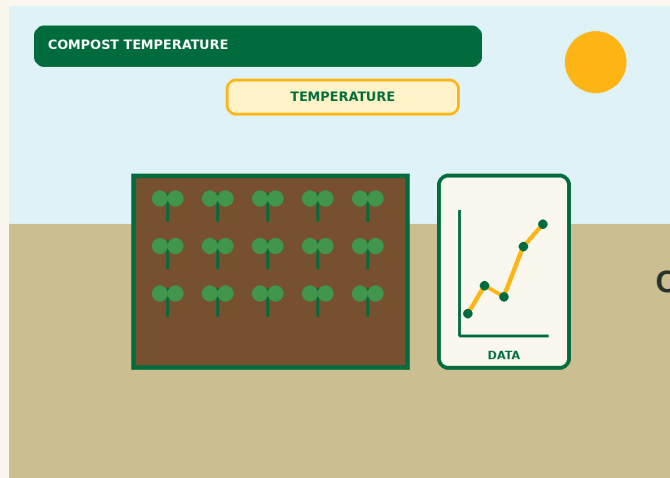
CONCEPT

Compost moisture must support microbial activity without filling too much pore space and restricting oxygen.

INVESTIGATION TASK

Evaluate a teacher-prepared moisture scenario.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TEMPERATURE

COMPOST TEMPERATURE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 727 • COMPOST TEMPERATURE

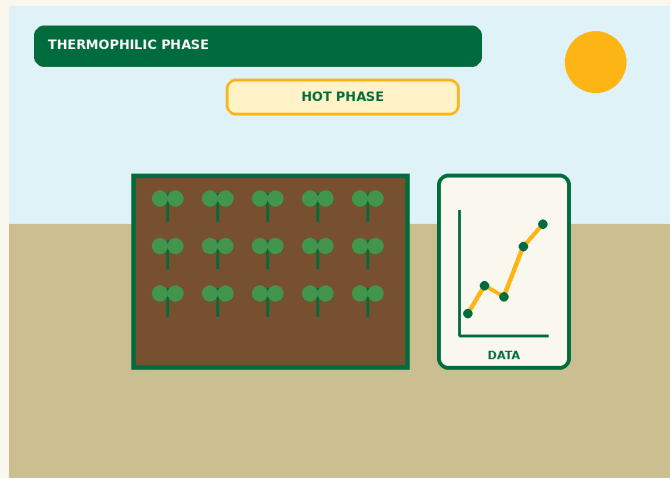
CONCEPT

Temperature patterns reflect biological activity, pile size, moisture, aeration, and stage of decomposition.

INVESTIGATION TASK

Graph a short temperature record.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HOT PHASE

THERMOPHILIC PHASE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 728 • THERMOPHILIC PHASE

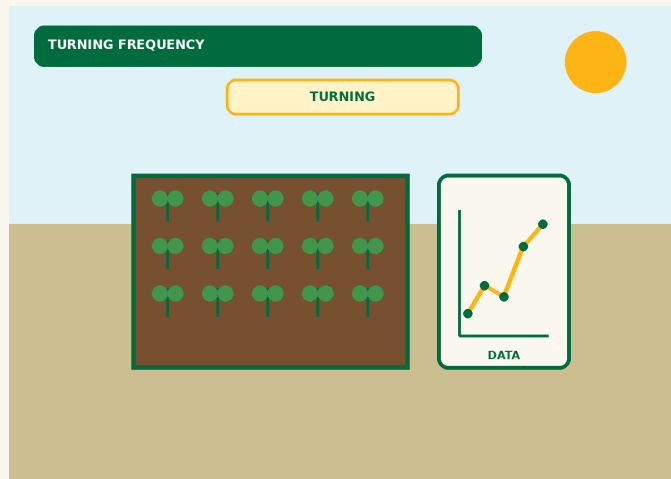
CONCEPT

The thermophilic phase is a high-temperature stage associated with rapid microbial activity under suitable conditions.

INVESTIGATION TASK

Identify it on a temperature-time graph.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TURNING

TURNING FREQUENCY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 729 • TURNING FREQUENCY

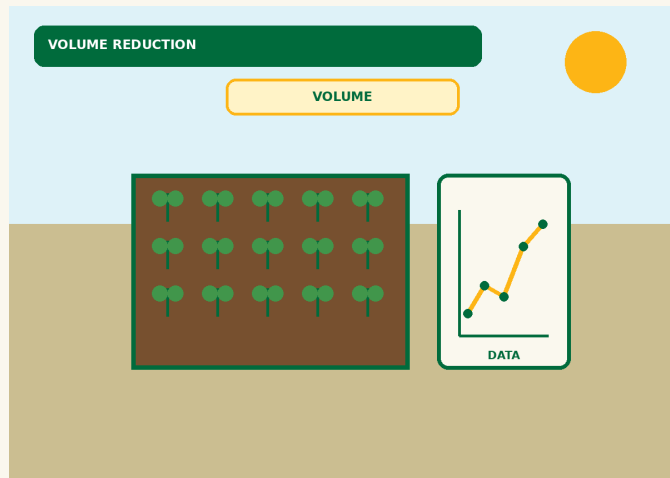
CONCEPT

Turning frequency affects aeration, mixing, moisture redistribution, and temperature patterns.

INVESTIGATION TASK

Compare two management schedules.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



VOLUME

VOLUME REDUCTION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 730 • VOLUME REDUCTION

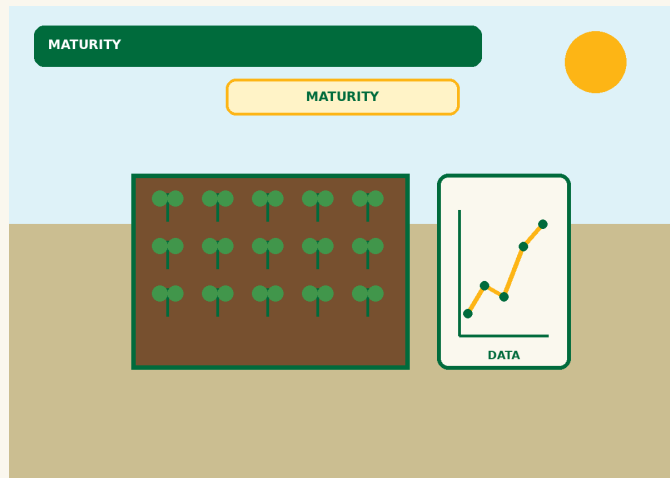
CONCEPT

Organic material volume commonly decreases as particles break down and carbon is transformed during composting.

INVESTIGATION TASK

Calculate percent volume reduction.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MATURITY

MATURITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 731 • MATURITY

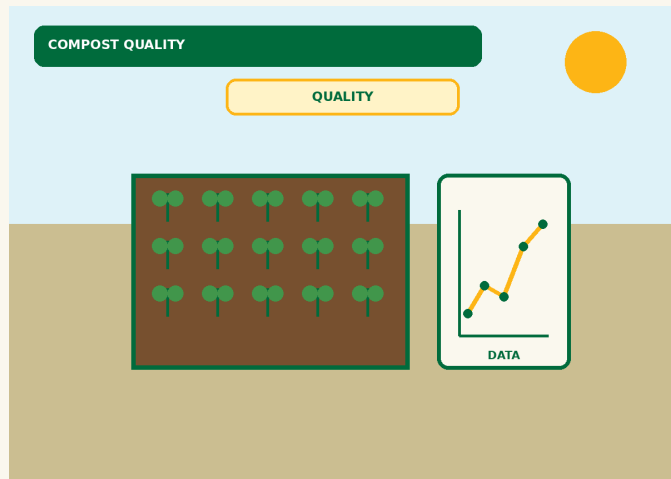
CONCEPT

Compost maturity indicates that active decomposition has slowed and the material has become more stable for appropriate use.

INVESTIGATION TASK

Compare characteristics of active and mature compost.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



QUALITY

COMPOST QUALITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 732 • COMPOST QUALITY

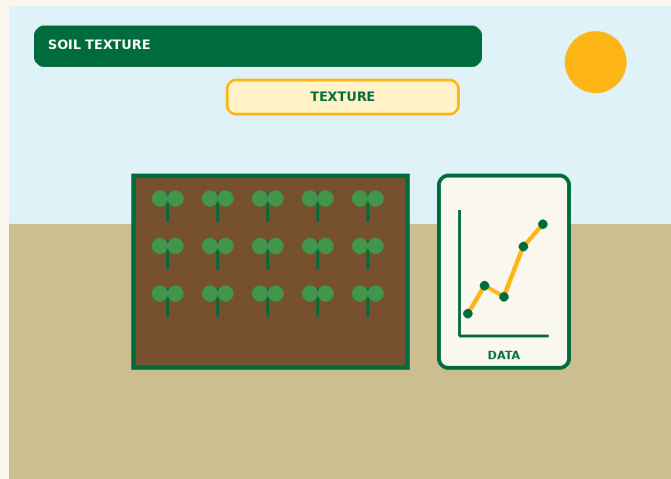
CONCEPT

Compost quality can be evaluated using maturity, contamination, particle condition, odor, moisture, and intended-use criteria.

INVESTIGATION TASK

Score a sample using a rubric.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TEXTURE

SOIL TEXTURE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 733 • SOIL TEXTURE

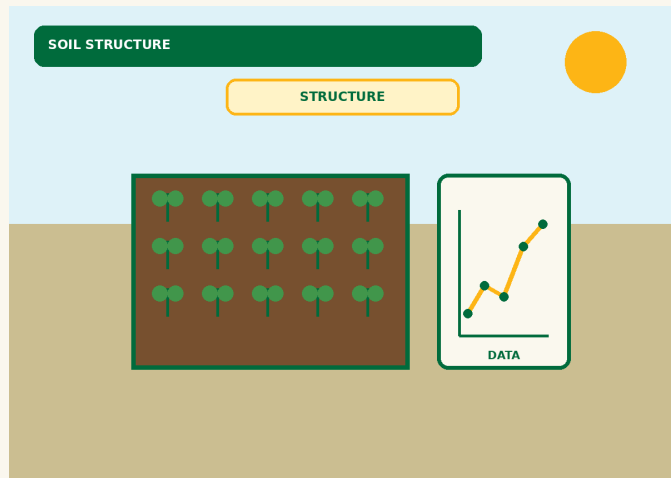
CONCEPT

Soil texture is determined by the relative proportions of sand, silt, and clay and influences water and nutrient behavior.

INVESTIGATION TASK

Classify teacher-provided texture information.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



STRUCTURE

SOIL STRUCTURE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 734 • SOIL STRUCTURE

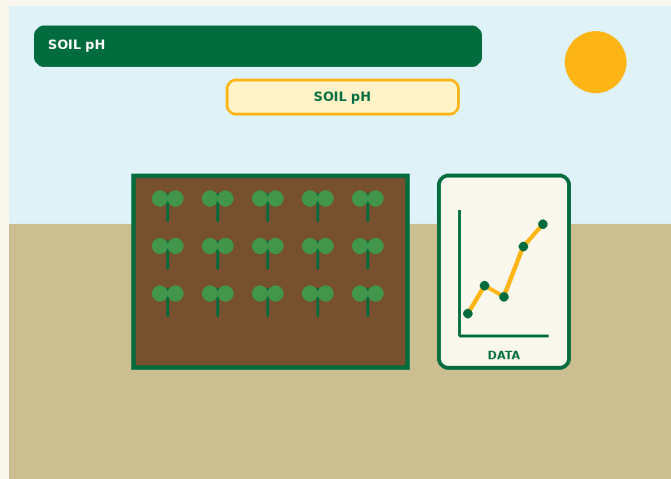
CONCEPT

Soil structure describes how particles form aggregates and pore networks that influence roots, water, and air.

INVESTIGATION TASK

Compare aggregate condition in two samples.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SOIL pH

SOIL pH

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 735 • SOIL pH

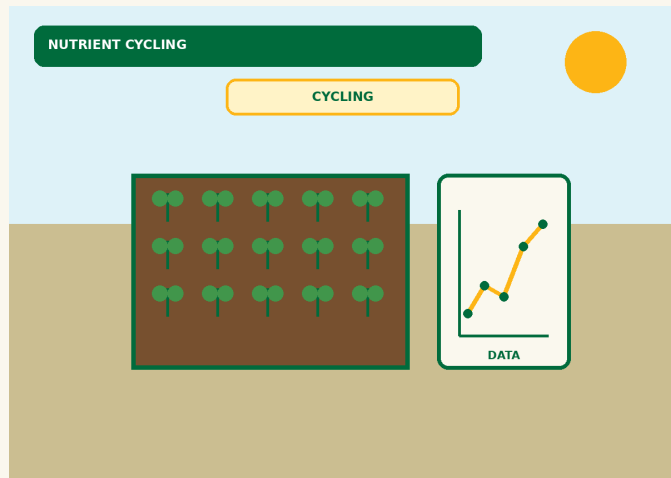
CONCEPT

Soil pH affects chemical reactions and nutrient availability and is measured on a logarithmic acidity-alkalinity scale.

INVESTIGATION TASK

Interpret a soil pH result.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CYCLING

NUTRIENT CYCLING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 736 • NUTRIENT CYCLING

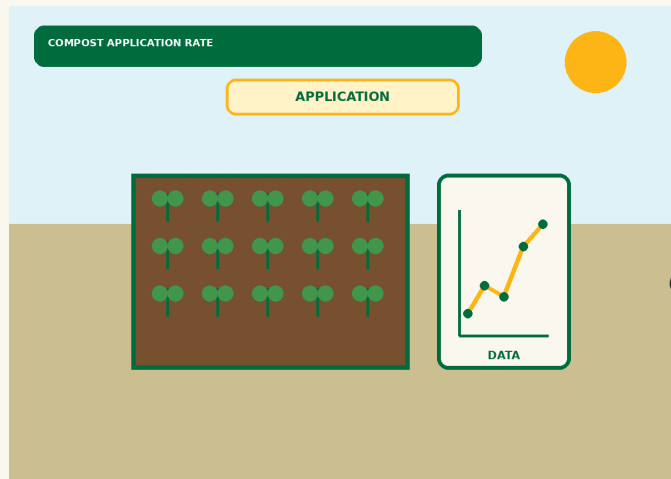
CONCEPT

Nutrient cycling moves elements among residues, organisms, soil organic matter, mineral forms, plants, and harvested products.

INVESTIGATION TASK

Draw one nutrient pathway.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



APPLICATION

COMPOST APPLICATION RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 737 • COMPOST APPLICATION RATE

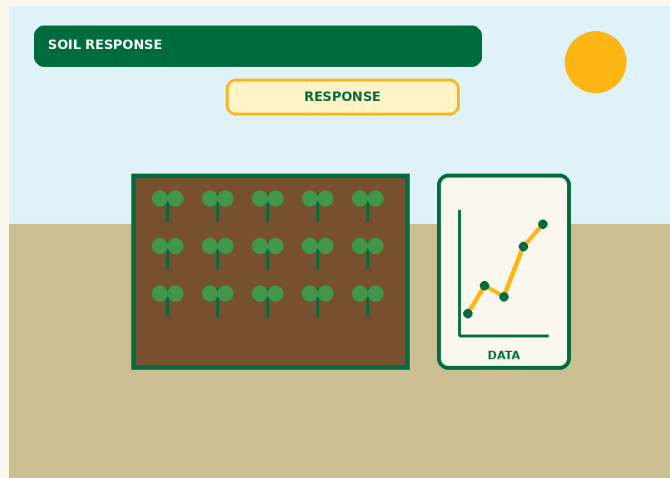
CONCEPT

Application rate expresses the mass or volume of compost added per unit area or soil volume.

INVESTIGATION TASK

Calculate a bed application quantity.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESPONSE

SOIL RESPONSE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 738 • SOIL RESPONSE

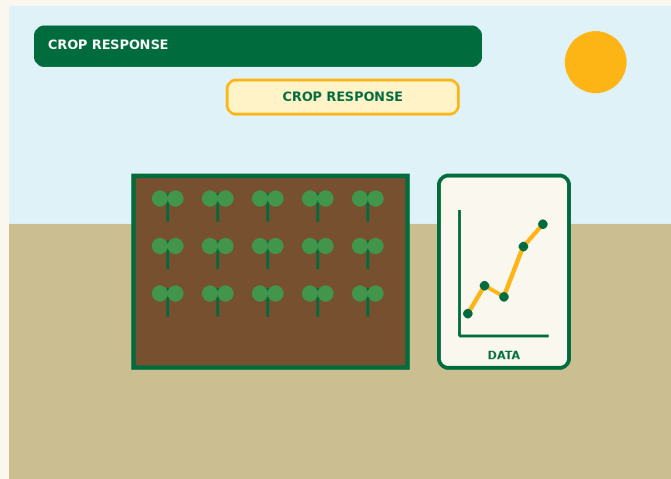
CONCEPT

Soil response to compost may include changes in organic matter, structure, nutrient supply, water behavior, and biological activity.

INVESTIGATION TASK

Design one measurable comparison.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CROP RESPONSE

CROP RESPONSE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 739 • CROP RESPONSE

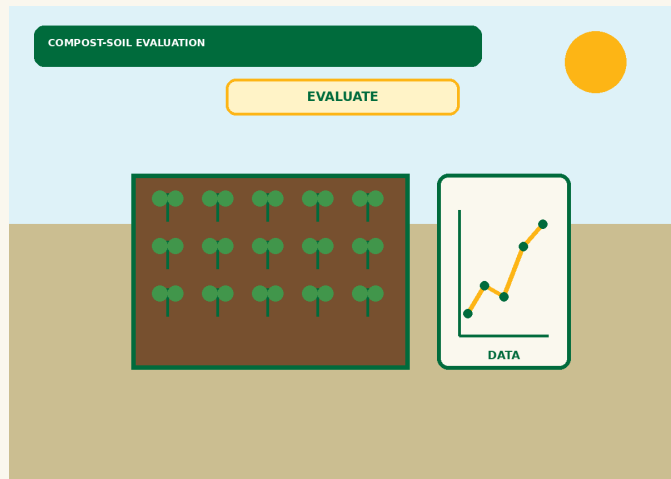
CONCEPT

Crop response can be evaluated using establishment, growth, condition, yield, and quality after soil management changes.

INVESTIGATION TASK

Select indicators for a compost trial.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EVALUATE

COMPOST-SOIL EVALUATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 740 • COMPOST-SOIL EVALUATION

CONCEPT

Evaluation integrates compost process records, quality, application rate, soil condition, and crop response.

INVESTIGATION TASK

Recommend an evidence-based compost-management adjustment.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE